

CPE 470/670 Fall 2025
Homework 3

Instructor: Parikshit Maini

Submission Deadline: 11:59 pm Monday, November 24, 2025

All text answers must be written using Latex. Paper submissions or scanned submissions will not be accepted. All code must be written in Python. Submit individual python files for every question. Zip together all submission material and submit the Zip folder. All questions are compulsory unless noted otherwise.

You are not permitted to seek help from people outside the course or AI based tools. You are permitted to refer to class notes, lecture materials, books and other online teaching resources.

If you do not have access to a printer, camera calibration targets are available in room 342 D in SEM building. This is a shared resource, do not remove it from the room. 342D is only available to students enrolled in CPE 470/670 to do course project related work. You can access the room during University business hours. Access code is 5791.

Question 1 (15 points) : You are provided with an ArUco marker (ArUco.png) from the DICT_4X4_50 dictionary pattern. Detect the ArUco marker using the related openCV functions in ArUco Marker Detection module. Use of *drawDetectedMarkers()* built-in function is not permitted. It is okay to use other functions in the library including *detectMarkers()*. Write Python code to detect the corners of the marker, create a boundary box around the ArUco marker by connecting its corners with green lines, and show the center of the ArUco marker by a red dot. Detect and display the ID of the marker at the left top of the image and save the annotated image. You can use related OpenCV functions to do these. Submit the annotated image along with your code. Your code will be run on other markers from the dictionary to verify correctness.

ArUco related resources: wikipedia, openCV documentation, Predefined ArUco Dictionaries.

Question 2(5 points) : Write a ROS2 package to capture and extract image frames from the USB camera given to you (as part of the course project kit). Plot the image frames. Your code must be well commented.

Question 3(10 points) : Write Python code to perform camera calibration to accurately estimate the intrinsic parameters of your camera. A Chessboard and a Charuco board pattern are provided in room 342D in SEM building (you can use either). Save the camera calibration matrix. You are permitted to use any library calls for camera calibration. Submit your code and the calibration matrix.

Camera calibration related resources: using checkerboard, using ChArUco